

Omar Olivares Urrutia

AI / Machine Learning Engineer — LLM Agents, Evaluation & RAG Systems



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SUMMARY

ML Engineer specializing in intelligent agents, LLM evaluation, and production RAG systems. Bridging research rigor with engineering excellence, from benchmarking 19 LLMs on SQL error detection (DataEd'26) to building high-growth open-source coding agents. Experienced in end-to-end ML lifecycles—data collection, fine-tuning, and scalable serving—with a focus on high-impact, user-facing AI products.

SKILLS

ML/AI Specializations

Intelligent Agents ·
LLM Evaluation (LLM-as-a-Judge) ·
RAG Systems ·
Fine-tuning (LoRA/QLoRA) ·
Agentic Workflows ·
Open-Source Development

Frameworks & Tools

PyTorch · Hugging Face ·
LangChain · OpenAI/Anthropic APIs ·
Weights & Biases · MLX

Infrastructure & Serving

FastAPI · Docker · Microservices ·
PostgreSQL ·
Vector DBs (Pinecone/ChromaDB) ·
AWS/GCP

Programming Languages

Python (expert) · SQL · TypeScript

WORK EXPERIENCE

Senior AI Engineer at Maxxa Software SpA

Mar 2026 - Current

📍 Santiago, Chile

Chilean fintech specializing in credit and surety guarantees. Building the production AI and LLM systems behind Maxxa's financial products on the TECH team.

- Architected and built the governance, evals, and observability layer for the company's production LLM systems
- Contributing to the internal LLM platform (core and enterprise editions) that powers AI features across the company's financial products
- Helped build the platform's Model Context Protocol (MCP) server, letting AI agents and internal teams plug company data and tools directly into their workflows

AI Engineer at Universidad de Talca

Sept 2025 - Mar 2026

Developing an AI-powered educational platform for SQL database learning using multi-agent LLM systems. Addressing limitations in traditional database education through real-time, granular feedback.

- Architecting a multi-agent system to provide real-time syntactic, semantic, and logical feedback on SQL queries
- Evaluating and benchmarking LLM agents (Grok, GPT-4o, Claude 3.5, Gemini, DeepSeek) for educational tasks

- Built evaluation suite benchmarking 19 LLMs on 300 SQL cases using LLM-as-a-Judge + human-annotated ground truth
- Co-authored peer-reviewed paper accepted at DataEd'26 validating the system's pedagogical impact

AI Engineer & Developer Relations at Kilo Code

Mar 2025 - May 2025

📍 San Francisco, CA / Amsterdam

Early team member (12-person founding team) of Kilo Code, a high-growth fork of the Cline/Roo-Code open-source coding agent. Joined from project inception to scaling phase.

- Collaborated with the founding team to launch one of the fastest-growing open-source coding agents
- Built OpenRouter integrations and technical documentation contributing to #1 model ranking on OpenRouter
- Developed integration guides and reference implementations for developer onboarding; project grew to ~10K GitHub stars
- Worked directly on technical architecture and strategy with leadership (GitLab founder Sid Sijbrandij's team)

AI Engineer at Emergent Mind

Feb 2024 - Feb 2025

📍 North Carolina, US (Remote)

Led AI development for an innovative arXiv research discovery platform, building production RAG systems to improve scientific search and analysis.

- Redesigned RAG pipeline (chunking + hybrid retrieval + reranking) improving search relevance for 200K+ indexed papers
- Built ingestion and indexing pipelines processing daily arXiv submissions; implemented evaluation harness with Weights & Biases
- Deployed FastAPI services with Docker; added tracing and reliability tests for production serving
- Conducted experiments to improve model performance and accuracy; oversaw QA processes for AI-driven features

Content Developer at Neura Pod

Sept 2021 - Jan 2025

📍 San Diego, CA (Remote)

Led content creation for a popular series exploring neurotechnology and AI, translating complex scientific concepts for a broad audience.

- Translated advanced research into engaging content, contributing to channel growth exceeding 80,000 YouTube subscribers
- Produced multiple viral videos surpassing 200,000 views each, fostering public understanding of BCIs and AI
- Managed end-to-end production: research, scripting, and distribution across major social media platforms

AI Evaluation Contractor at Centro Nacional de Inteligencia Artificial

Nov 2024 - Nov 2024

📍 Santiago, Chile

Participated in rigorous AI model evaluation tasks for LATAM Spanish language models.

Software Engineer at clicOH

Jul 2021 - Dec 2021

📍 Remote

Software Engineer at a logistics startup (originally RayoApp, acquired by clicOH).

- Contributed to backend development and system integration post-acquisition
- Maintained legacy systems while migrating features to new architecture

Software Engineering Intern at Suncast

Jan 2020 - Apr 2020

📍 Las Condes, Chile

Summer internship during Computer Science studies. Helped rebuild the platform into a scalable microservices-based architecture.

- Refactored monolithic application into microservices to improve scalability and maintainability
- Implemented API endpoints and optimized database queries for renewable energy forecasting platform

Information Technology Support Assistant at Universidad de Talca

Jan 2016 - Dec 2018

📍 Curicó, Chile

Responsible for network resolution, technical support, and IT infrastructure deployment.

- Solved network issues and deployed IT infrastructure alongside the technical team

Music Producer at Ableton

Jan 2008 - Dec 2013

📍 Valencia, Spain

Composed, produced, and engineered tracks for artists.

PROJECTS

Graham Essays

Jul 2022 - Current

Skills: Python · Web Scraping · Open Source

🔗 <https://github.com/ofou/graham-essays>

Automated EPUB generator for Paul Graham's essays — reached #1 on Hacker News, 800+ GitHub stars

- Reached #1 on Hacker News frontpage
- Built automated pipeline processing 200+ essays into EPUB format
- 800+ GitHub stars; demonstrates open-source community engagement

Lucas — AI Finance Agent

Nov 2025 - Current

Skills: Anthropic API · Agents · FastAPI

🔗 <https://github.com/platanus-hack/platanus-hack-25-team-29>

Agentic AI personal finance assistant with real-time bank API integration

- Built AI-powered finance agent connecting to bank accounts via Fintoc API
- Developed within 36-hour hackathon using Anthropic Claude API + FastAPI backend

VOLUNTEER

Software Engineer at Neuromatch Academy

Jul 2021 - Sept 2021

Neuromatch is an online community of computational neuroscientists whose mission is to foster inclusive global interactions for learning, mentorship, networking, and professional development.

EDUCATION

Postgraduate Diploma (Feb 2026) Generative AI and Large Language Models at Centro Nacional de Inteligencia Artificial (CENIA)

Aug 2025 - Feb 2026

Major courses: LLM Fine-tuning (LoRA, QLoRA, PEFT) · Production ML (PyTorch, MLflow, W&B) · Multimodal Models & RAG Systems · Ethical AI & Bias Management

Engineer's degree Computer Science at Universidad Católica del Maule

Mar 2018 - Dec 2023

Major courses:

Thesis: Fine-tuned GPT-3.5 for Chile's PAES math exam — improved accuracy from 0.78 to 0.94 (+20%) ·

Data Structures & Algorithms, Database Systems, Distributed Systems, AI/ML

Engineer's degree Computer Engineering at Universidad de Talca

Mar 2015 - Dec 2017

AWARDS

2nd Place — Computer Vision Hackathon at Fundación País Digital | EY

Sept 2025

Built traffic monitoring system detecting congestion and anomalies from CCTV streams, scalable to thousands of cameras.

CERTIFICATES

IELTS Academic - Overall Band Score 7.5 (C1 Level) Cambridge English

Jul 2025

PUBLICATIONS

Assessing the Ability of Large Language Models to Identify and Explain Errors in SQL Queries in DataEd'26: International Workshop on Data Systems Education

Mar 2026

Empirical evaluation of 19 LLMs on SQL error detection. Used LLM-as-a-Judge paradigm (300 test cases) and human-annotated ground truth. Findings highlight LLM potential for automated tutoring systems.

Optimization of Language Models for Chile's Mathematics Higher Education Admission Test (PAES) in Universidad Católica del Maule

Feb 2024

Fine-tuned GPT-3.5 for Chile's PAES mathematics exam. Improved accuracy from 0.78 to 0.94 (+20%) through targeted optimization techniques and OCR/transcription pipelines.

LANGUAGES

Spanish

Native

English

Full Professional (IELTS C1 7.5)

INTERESTS

Technology

Skills: Neurotechnology · Brain-machine interfaces · Open source

Art

Skills: Music production · Film making · Technical writing

Languages

Skills: Learning Mandarin

Travel

Skills: Cultural explorations · Adventure travel · Culinary experiences

REFERENCES

“ Co-founder, Kilo Code ”

— Matt Mazur

“ Founder, Emergent Mind ”

